

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO2	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
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Section/Batch	Slot C /	Date	03-10-2026

Code of Conduct:

I, _____, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

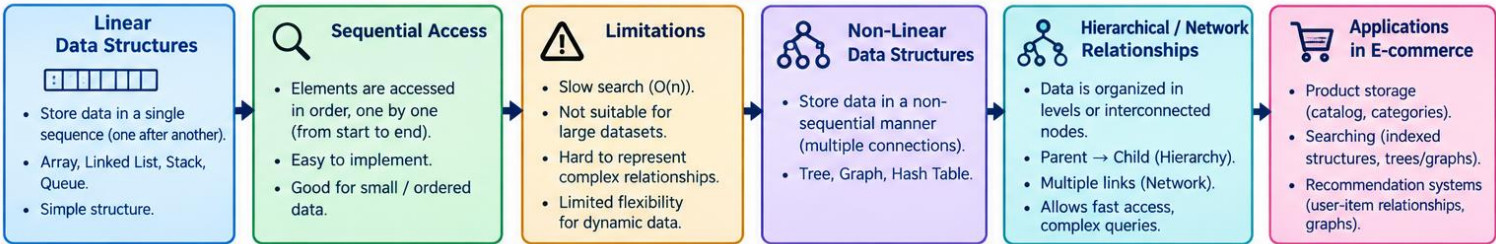
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Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

Concept Map

E-commerce Platform – Data Structures

Managing product storage • Searching • Recommendation systems



Linear vs Non-Linear Data Structures – Key Differences		
Feature	Linear Structures	Non-Linear Structures
Structure	Single sequence	Multiple levels / connections
Access	Sequential (one by one)	Direct / multiple paths
Search Time	$O(n)$ – slower	$O(\log n)$ or $O(1)$ (faster)
Complexity	Simple	More complex
Relationship Representation	Limited	Supports hierarchical and network relationships
Best Use	Small, ordered data, simple operations	Large data, complex queries, relationships

How Each Structure Supports E-commerce System Components	
Linear Structures	Non-Linear Structures
Product Storage Array / Linked List to store product details (ID, name, price, stock).	Product Storage Tree (category hierarchy) or Graph (brand–category–product).
Searching Sequential search (simple / small catalog) or sorted array (faster lookup).	Searching Balanced Tree (faster search) or Hash Table (quick lookup by ID).
Recommendation Stack / Queue (e.g., recent views, last added products).	Recommendation Graph (user-item relationships, collaborative filtering).

Which Structure is More Efficient for Complex Applications?



- Non-linear data structures are more efficient for complex e-commerce applications because:**
 - They can represent hierarchical data (categories, subcategories) and network relationships (users, products, recommendations).
 - They provide faster search and retrieval using trees, graphs or hash tables.
 - They handle large-scale data and dynamic relationships better than linear structures.
- So, a combination is ideal: Linear structures for simple and ordered data, and non-linear structures for complex, scalable, and relationship-based features like search and recommendation systems.